

NOXTILL — Master Build Prompt (Complete Backend + Frontend + SEO + Analytics)

Copy everything below this line and give it to your development team or AI coding tool. This is the complete, final specification.

Build **Noxtill** — a production-ready, multi-tenant, global SaaS platform that is the complete "Business Operating System" for small and medium businesses worldwide. Tagline: **"Close your day in one tap."** Brand story: *nox* (Latin: night) + *till* (the shop counter) — every night, Noxtill closes the owner's books automatically and sends them their whole day in one message. Noxtill unifies in one system what businesses today juggle across six disconnected apps: review and reputation management, bookings and appointments, orders and billing, a digital customer credit ledger, inventory, profit and loss analytics, customer CRM, staff management, marketing, deep search, business analytics, and AI assistance. It serves every customer-facing business type on earth — restaurants, cafes, bakeries, clinics, dentists, salons, barbershops, gyms, lawyers, accountants, retail shops, e-commerce stores, plumbers, electricians, hotels, schools, photographers, agencies, software companies, wholesalers, and any business not on this list via an AI-generated custom setup. The defining product principle is **automation through connection**: every module triggers the next with zero owner effort — a booking sends its own confirmation and reminders; a completed visit triggers a review request; a sale updates stock, profit, and the customer record simultaneously; and every night at a configurable local time (default 10 PM in the business's timezone) the owner receives the **Nightly Close** — Noxtill's signature feature — a single message on their preferred channel summarizing the day's sales, profit, new reviews, tomorrow's bookings, credit recovered, and low-stock alerts, with a one-tap link to the full report.

Positioning: beat the incumbents by being one system, not six

Noxtill must be demonstrably better than the fragmented alternatives a business would otherwise assemble: reputation platforms (Podium, Birdeye) that handle reviews but not sales; POS systems (Square, Lightspeed) that handle sales but not reputation; booking tools (Calendly, Fresha) that handle appointments but nothing after them; accounting suites (Zoho, Odoo, QuickBooks) that are powerful but too complex for a shop owner and blind to WhatsApp; and credit-ledger apps (Khatabook-style) that do one thing only. Noxtill wins on four axes that must guide every implementation decision: **integration** (one login, one customer record, one report — data flows between modules automatically, which no point-solution can offer); **simplicity** (a non-technical shop owner must be productive in ten minutes; the three-second dashboard rule below is law); **messaging-first delivery** (the product lives where owners live — WhatsApp, SMS, email — not buried in a web app they

forget to open); and **honest AI** (AI that works on the owner's own data and never fabricates). The following capabilities are Noxtill's unique differentiators that no single competitor offers together, and every one of them must ship: the Nightly Close report; the built-in customer credit ledger with polite automated reminders; the AI-generated dashboard for any business type on earth; the review-to-revenue loop (reviews, bookings, and sales in one funnel); the AI Business Assistant reachable in-app and later over WhatsApp; the what-if profit calculator; full offline mode; per-business white-label customer pages; and the growth engine described in the SEO module that turns every customer into a marketing surface.

Global-first Requirements (apply to every module)

Everything is international from day one and nothing regional is hard-coded — currency, language, tax, payment provider, and message channel are all configuration. **Multi-currency:** every business selects its ISO 4217 currency at signup; all money fields, reports, invoices, and AI answers use the business's currency and number formatting. **Multi-language:** the entire interface is internationalized with all strings in translation files, launching in English, Spanish, Portuguese, French, Arabic, Urdu, and Hindi, with full right-to-left layout support and automatic AI language matching (an Urdu review gets an Urdu reply; a Spanish owner gets Spanish reports). **Multi-timezone:** all scheduling — reminders, follow-ups, the Nightly Close — executes in the business's local time. **Multi-region messaging:** WhatsApp Business API via Meta's Cloud API directly (no BSP fees) where WhatsApp dominates, Twilio SMS with pluggable local gateways elsewhere, and transactional email via SES or Postmark, with per-region channel defaults and per-business overrides; transactional messages (confirmations, reminders, receipts, credit reminders) are classified under WhatsApp's cheaper Utility category, promotional content under Marketing, and the free 24-hour service window after a customer reply is used for conversational flows. **Multi-region payments:** Stripe as the global default with a pluggable provider architecture for regional gateways (JazzCash/Easypaisa, Razorpay, Mercado Pago, Paystack) added per market without core-code changes. **Tax:** configurable rates and labels (VAT, GST, sales tax) on invoices and reports. **Privacy:** GDPR-grade baseline everywhere — tenant isolation, marketing consent tracking, opt-outs honored across channels, right to erasure, and full data export.

Technology Stack and Architecture

Backend: **Node.js with NestJS**. Database: **PostgreSQL**, strictly multi-tenant — every table carries `business_id`, every query is tenant-scoped at the data-access layer automatically, and cross-tenant access is impossible at every layer including exports, webhooks, search indexes, and AI tool calls. Queues: **Redis with BullMQ** — no external call (messaging, Google, payments, AI, PDF generation) ever runs synchronously inside a user request; every external call is a queued job with retries, exponential backoff, a dead-letter queue, and per-message delivery-status tracking (sent, delivered, read, failed) visible in the dashboard.

Web frontend: **Next.js + React + Tailwind CSS** with React Query. Mobile: **React Native**, one codebase for iOS and Android. Search: **PostgreSQL full-text search initially, upgradable to Meilisearch/Typesense** for the Deep Search module. All webhooks (Meta, Google, Stripe, regional gateways) are **idempotent**: every event carries a stored unique ID and repeats are acknowledged and ignored. An **immutable audit log** records every financially significant action (sales, edits, payments, credit entries, refunds, deletions, exports) with actor, business, before/after values, and timestamp; audit rows can never be modified. **Role-based access control**: Owner (everything), Manager (operations, not billing or full exports), Staff (own sales and assigned tasks only; no customer-database export). Generated files live in S3-compatible storage behind **signed URLs expiring in 24 hours**. Daily automated database backups with tested restores. AI: the **Anthropic Claude API**, using a fast low-cost model for high-volume tasks. Hosting starts on a single cloud provider with CI/CD, monitoring, and error tracking from week one.

Module 1 — Reviews and Reputation

After a completed sale, order, or appointment, a review request is automatically enqueued after a configurable delay (default two hours) and sent on the business's regional channel in the customer's language, with gentle automatic reminders at three and seven days. The customer lands on a mobile-first, white-label rating page (business logo, colors, name) and picks one to five stars. Routing is platform-policy compliant with **no review gating**: everyone gets the same link; four-to-five-star customers get a one-tap redirect to the business's public review profile (Google by default; configurable to Yelp, Facebook, Trustpilot, TripAdvisor per market); one-to-three-star customers are offered a private feedback form that alerts the owner instantly and never posts publicly. The dashboard unifies reviews from Google (free Business Profile API), Facebook, and other connected platforms in one inbox with in-dashboard replies, converts private complaints into assignable, trackable tickets, and charts rating trends, request-to-review conversion, and complaint themes. Every business gets an auto-generated **QR code** with print-ready counter-stand designs linking to its rating page, and an embeddable **review widget** showing its best reviews on its own website with a "Powered by Noxtill" backlink.

Module 2 — Bookings and Appointments

Each business gets a public booking link and QR where customers pick a service (duration, price), optionally a staff member, and a slot computed from working hours, staff schedules, and existing bookings, with optional advance deposits via the connected payment provider. The business sees a day/week calendar with staff columns, drag-to-reschedule, walk-in entry, and statuses (booked, confirmed, completed, no-show, cancelled). Automation: instant confirmation; reminders at configurable intervals (default 24 hours and 2 hours) containing one-tap reschedule/cancel links that update the calendar; completion fires the Module 1 review request; no-shows are recorded on the customer profile.

Module 3 — Products, Orders, and Billing (POS-lite)

Catalog: products and services with name, photo, category, variations, **cost price**, **selling price**, stock quantity, service duration and rate, and CSV bulk import. Orders are created at the counter, via a public online-order link, or by converting an approved quotation, with statuses pending → confirmed → in progress → delivered/completed → refunded, and automatic customer status messages. Restaurant mode adds dine-in with table numbers and a table-status board, takeaway, delivery, and a QR table menu. E-commerce mode adds shipping and payment-status fields. Billing: a fast one-tap sale screen with discounts and tax; branded PDF invoices and receipts printable on standard receipt printers or sent by message; quotation PDFs that convert to jobs on approval; payments across cash, card, connected online providers, and **credit** (Module 4); barcode-scanner support. Every sale automatically decrements stock, computes per-line profit, appends to the customer's CRM history, and flows into the Nightly Close and the P&L engine.

Module 4 — Customer Credit Ledger

Digitizes the informal credit ("tab", "udhaar", "khata", "fiado") that small businesses worldwide extend to regulars — a flagship differentiator incumbents ignore. A credit-type sale creates or updates a per-customer ledger keyed by phone number with running balance, dated entries, and payments. The credit screen lists every debtor and total receivable, generates per-customer PDF statements, and sends **one-tap polite reminders** to one or all debtors. All entries are audit-logged; recoveries appear in the Nightly Close; the UI label localizes per market.

Module 5 — Inventory

Stock auto-decrements on sale and increments on purchase entries (supplier, quantity, cost — keeping cost prices current). Per-product low-stock thresholds trigger dashboard and Nightly Close alerts. Wastage tracking records discarded stock with reasons so true profit stays accurate. Inventory value at cost is always visible.

Module 6 — Profit and Loss Engine

Per-product margins and currency profit per unit with explicit loss-maker flags; hourly/daily/weekly/monthly sales-time analysis ("60% of your sales happen between 7 and 10 PM"); best sellers and dead stock; manual and recurring expense entries (rent, salaries, utilities); and a true monthly P&L (revenue – cost of goods – expenses) exportable for accountants and banks. Includes the **AI what-if calculator** answering hypotheticals ("what if I raise this price 10%?") strictly from the business's own history, always labeled as an estimate, never a guarantee.

Module 7 — Customer CRM

Every customer record is keyed by phone number and built automatically with zero data entry: name, contact details, optional birthday; full purchase and visit history with serving staff; lifetime spend, visit count, last visit, average return interval; credit balance; reviews and complaints; automatic rule-based tags (VIP by spend threshold, New, Lapsed by absence) plus manual tags; and free-text owner notes. Tools: instant search by name or partial number; one-tap message from any profile; one-tap segment campaigns (lapsed win-backs, VIP offers); automatic birthday greetings; per-customer consent and opt-out status honored everywhere; per-customer erasure on request. Each business's customer data belongs to that business alone — never sold, never shared, never messaged by anyone else.

Module 8 — Staff Management

Staff accounts with roles; attendance check-in/out; sale and appointment attribution; configurable per-staff commission rules with automatic monthly commission reports; performance dashboards; and a team inbox where complaints and tasks are assigned and resolved.

Module 9 — Marketing and Growth Tools

One-click broadcast campaigns to all customers or CRM segments with quotas, rate limiting, and opt-out compliance; automatic social posts turning five-star reviews into branded images on connected Facebook/Instagram pages; a referral program ("give a friend 10%, get 10%") with trackable codes sent after positive reviews; competitor tracking (the owner adds three to five local rivals; weekly rating/review-count comparisons plus a local-search ranking indicator); and the QR assets and review widget doubling as marketing surfaces.

Module 10 — Multi-location and Enterprise

One account, many branches: per-branch dashboards, staff, and catalog overrides; a combined roll-up; side-by-side branch comparisons; and an **AI branch advisor** that explains performance gaps between the owner's own branches from their own data only — it never advises on external questions (like where to open next) for which it has no reliable data.

Module 11 — Customizable Widget Dashboard and Universal Business-Type System

The dashboard is a grid of independent **widgets** (today's sales, today's profit, new reviews, average rating, today's bookings, pending orders, active tables, occupancy, hearings today, fees due, members present, low stock, credit outstanding, staff commissions, top product,

and more — twenty-five to thirty at launch, growing with every module), backed by three structures: a `widgets` registry, `dashboard_templates` mapping each business category to a default layout, and per-business `dashboard_config` JSON storing the owner's arrangement. Universal coverage uses the **three-level system**: Level 1 — roughly twelve base patterns (food and dining; health and appointments; beauty and personal care; fitness and memberships; professional services; retail and e-commerce; field services; hospitality and rentals; education; software and agencies; manufacturing and wholesale; general). Level 2 — two to three hundred named business types beneath them adjusting labels and default widgets; adding one is a database row, never a deployment, and each type powers its own SEO landing page (see SEO module). Level 3 — the **AI fallback**: an unlisted business describes itself in one line; Claude returns structured JSON mapping it to the best pattern, widgets, and labels; the dashboard is generated and the new type is saved, so the directory grows itself worldwide. Home-screen law: the **three-second rule** — big numbers, prominent alerts, the day understood at a glance — and every number clicks through to its underlying list. Launch with templates; add drag-and-drop customization (dnd-kit or react-grid-layout) with an add-widget gallery in a later phase.

Module 12 — Deep Search (global in-app search)

A single search bar, available on every screen and via keyboard shortcut, that searches **everything in the business's own data at once**: customers (name, partial phone), orders and invoices (number, amount, item), products, bookings, reviews and complaints, credit entries, staff, expenses, and even settings and help articles — with typo tolerance, instant-as-you-type results grouped by category, and deep links straight to the record.

Implementation: PostgreSQL full-text search at launch, upgradable to Meilisearch/Typesense behind the same interface; the index is strictly tenant-scoped. Deep Search also powers the AI Assistant's retrieval and the CRM's instant lookup. The goal: an owner never navigates menus to find anything — they type three characters and land on it.

Module 13 — Analytics (product analytics for owners + platform analytics for Noxtill)

For business owners: an Analytics section unifying every module's numbers — revenue, profit, orders, bookings, review scores, customer growth, retention cohorts (how many of last month's customers returned), campaign performance, staff performance, and channel stats (message delivery/read rates) — over any date range, with comparisons (this month versus last), all clickable to underlying records, all exportable, and all summarized monthly by the AI in plain language. **For the Noxtill team**: built-in event instrumentation (signups, activation steps, feature usage, message volume, plan upgrades, churn signals) feeding an internal admin panel with tenant lists, MRR, churn, cohort retention, and usage, so the company runs on its own data from day one. Every metric shown to an owner must be traceable to source records — no vanity numbers.

Module 14 — AI Assistant

Two strictly separated capabilities in one chat surface. **Help Assistant:** answers "how do I..." questions in the user's language via retrieval-augmented generation over Noxtill's own documentation, with deep links to the right screens — it can never invent features.

Business Assistant: answers questions about the owner's own data ("today's sales?", "what sold most this week?", "what's this customer's balance?") via Claude **tool-calling** over fifteen to twenty read-only backend functions (get_todays_sales, get_top_products, get_customer_balance, get_low_stock_items, get_staff_performance, get_review_summary, and similar), each executed server-side hard-scoped to the authenticated business_id. Hard rules: read-only in version one; every quoted number comes from a tool result; when tools return nothing the assistant says so and points to the right screen instead of guessing; one genuinely useful proactive note is welcome (a low-stock warning after a top-products answer). Later phase: the same assistant over WhatsApp/SMS, so an owner messages their own system "today's sales?" and gets the answer without logging in — a capability no incumbent offers.

Module 15 — Reports, Downloads, and Data Export

Every relevant screen has a download action and every generated file has a **"send by message"** button. Branded PDF reports via HTML templates rendered with Puppeteer (business language and currency): monthly business report with AI insights; P&L statement; sales over any range; staff performance and commissions; review/reputation reports. Excel/CSV via exceljs: sales, customers, credit balances, stock, expenses. Operational documents: invoices, receipts, quotations, QR posters, credit statements. Large exports run as queued background jobs delivering a signed 24-hour link. Settings offers a **full account export** as a zip — the explicit "your data is never hostage" promise. Export permissions follow roles (full customer export is owner-only).

Module 16 — Mobile App and Offline Mode

React Native owner app: push notifications for reviews, complaints, orders, bookings; today's numbers; quick sale entry; customer lookup; the AI Assistant. **Offline mode:** sales entered without connectivity queue locally and sync with conflict handling when the connection returns — essential for markets with unreliable power and internet, and absent from most Western competitors.

Module 17 — SEO and Growth Engine (complete SEO for everything)

SEO is a product module, not an afterthought, operating on three layers. **Layer 1 — Noxtill's own acquisition SEO:** the marketing site is server-rendered (Next.js SSR/SSG) with full technical SEO — clean semantic HTML, per-page unique titles and meta descriptions, Open Graph and Twitter cards, canonical URLs, XML sitemaps auto-updated, robots.txt,

schema.org structured data (Organization, SoftwareApplication, Product with pricing, FAQPage, BreadcrumbList), hreflang tags across all launch languages, Core Web Vitals budgets enforced in CI (LCP under 2.5s, CLS under 0.1), image optimization with alt text, and mobile-first rendering. On top of this build **programmatic SEO**: an automatically generated, genuinely useful landing page for every one of the two-to-three-hundred business types ("bakery management software", "clinic booking and billing system", "salon POS with reviews") and, in later phases, type-by-country pages ("salon software UK"), each with tailored copy, the relevant widgets shown, an interactive demo, FAQs with FAQ schema, and internal links — powered by the same business-type registry as Module 11 so new types create new pages automatically. Add a content/blog engine (MDX-based) for guides targeting owner problems ("how to reduce no-shows", "how to collect unpaid customer credit politely"), each with article schema and internal links to the relevant type pages. **Layer 2 — customers' SEO (product value that markets itself)**: every review collected through Noxtill improves the customer's own local search ranking; the review widget adds fresh review content (with Review/AggregateRating schema) plus a "Powered by Noxtill" backlink to their site; optional hosted business mini-pages (menu/services/booking) are schema-marked (LocalBusiness, Menu, Service) and fast, further boosting the customer's visibility — making Noxtill's pitch "we make Google love your business" literal. **Layer 3 — measurement**: integrate Search Console and analytics into an internal SEO dashboard tracking rankings, indexed pages, and signup attribution by landing page, so the programmatic pages are managed by data. All SEO work must be white-hat only — no gating, no fake reviews, no doorway spam; the programmatic pages must each contain genuinely distinct, useful content.

Subscription Plans (Noxtill's revenue)

Four self-serve monthly tiers, 14-day free trial, no card required, USD reference pricing localized per market: **Basic (\$10)**: reviews, routing, unified inbox, AI replies, QR, widget, 150 messages, 1 user. **Growth (\$25)**: plus bookings, orders, billing, invoices, credit ledger, CRM — 600 messages, 3 users. **Business (\$50)**: plus P&L engine, inventory, staff and commissions, AI Assistant, what-if calculator, analytics, reports/exports — 2,000 messages, 10 users. **Enterprise (from \$100)**: multi-location, branch AI, priority support, unlimited users, custom limits. Enforce message quotas with upgrade prompts; accept Stripe plus regional methods; run the internal admin panel from Module 13.

Brand and Frontend Design Direction

Identity: the world of the shop ledger and the receipt roll, closed each night under the moon — never a generic SaaS template. Tokens: warm paper background (□ #FAF7F0); deep ledger green (■ #0C4B3B) primary; ink (■ #182420) text; marigold (■ #E8A93C) accent; muted red (■ #B94A3D) reserved for outstanding credit. Type: Bricolage Grotesque display, Instrument Sans body, locale script fonts (Noto Nastaliq Urdu, Noto Naskh Arabic) as accents in localized versions. Motifs: ledger-ruled dividers, 14px-radius cards with subtle hover lifts, a

crescent-moon-plus-ledger-line logo mark for **Noxtill**. The landing hero pairs "Your whole business, one system" with the signature moment: a phone mockup of the Nightly Close arriving as a message, captioned "**Close your day in one tap.**" Page sections: stats strip; four-step how-it-works (booking → sale → review → Nightly Close); module grid covering every module above; credit-ledger feature band; AI section with sample assistant chat; business-types chip cloud with the AI custom promise; Deep Search demo; four pricing cards with Growth highlighted; trust section (data isolation, daily backups, full export, offline mode, speaks your language); final CTA ("get your first Nightly Close tonight"); footer. The app dashboard uses the same tokens, is fully responsive with a collapsing mobile sidebar, and meets accessibility baselines: visible keyboard focus, reduced motion respected, RTL correctness, adequate contrast.

Non-negotiable Rules

Never implement review gating; comply with every connected platform's policies. Never allow cross-tenant data access at any layer, including search indexes and AI tool calls. AI never writes or deletes data in version one and never states numbers that don't come from tool results. Never message opted-out customers; honor erasure requests. Hard-code nothing regional. Audit-log every financial mutation. Queue every external call. Make every webhook idempotent. Back up daily and test restores. All SEO is white-hat.

Three-Month Delivery Plan (aggressive, launch-first)

Month 1 — Foundation and the core loop: full database schema designed for all modules; multi-tenant architecture, auth, roles; messaging engine with queues and delivery tracking; WhatsApp and Google API approvals filed on day one; Reviews (Module 1) and Bookings (Module 2) complete end-to-end including rating pages, unified inbox, QR, calendar, and the first version of the Nightly Close; marketing site live with full technical SEO and the first fifty programmatic type pages. **Month 2 — Commerce and money:** Products, Orders, Billing (Module 3); Credit Ledger (Module 4); CRM (Module 7); invoices and receipts; Stripe billing with all four plans; Deep Search v1; beta cohort of twenty to thirty real businesses onboarded and giving feedback; launch and start charging at the end of this month. **Month 3 — Intelligence and polish:** Inventory (5), P&L engine with what-if (6), Staff (8), Marketing tools (9), Analytics (13), AI Assistant (14), Reports and exports (15); mobile app beta with offline mode; remaining programmatic SEO pages and the blog engine; hardening, load testing, security review, and the public launch push. Multi-location (10) and dashboard drag-and-drop customization follow in month four-plus. Non-negotiable sequencing rules: the schema is designed for everything in week one even though modules ship incrementally; the product is live and sellable from the end of month two; and if any phase slips, cut scope from the end of month three — never from quality, security, or the core loop.